

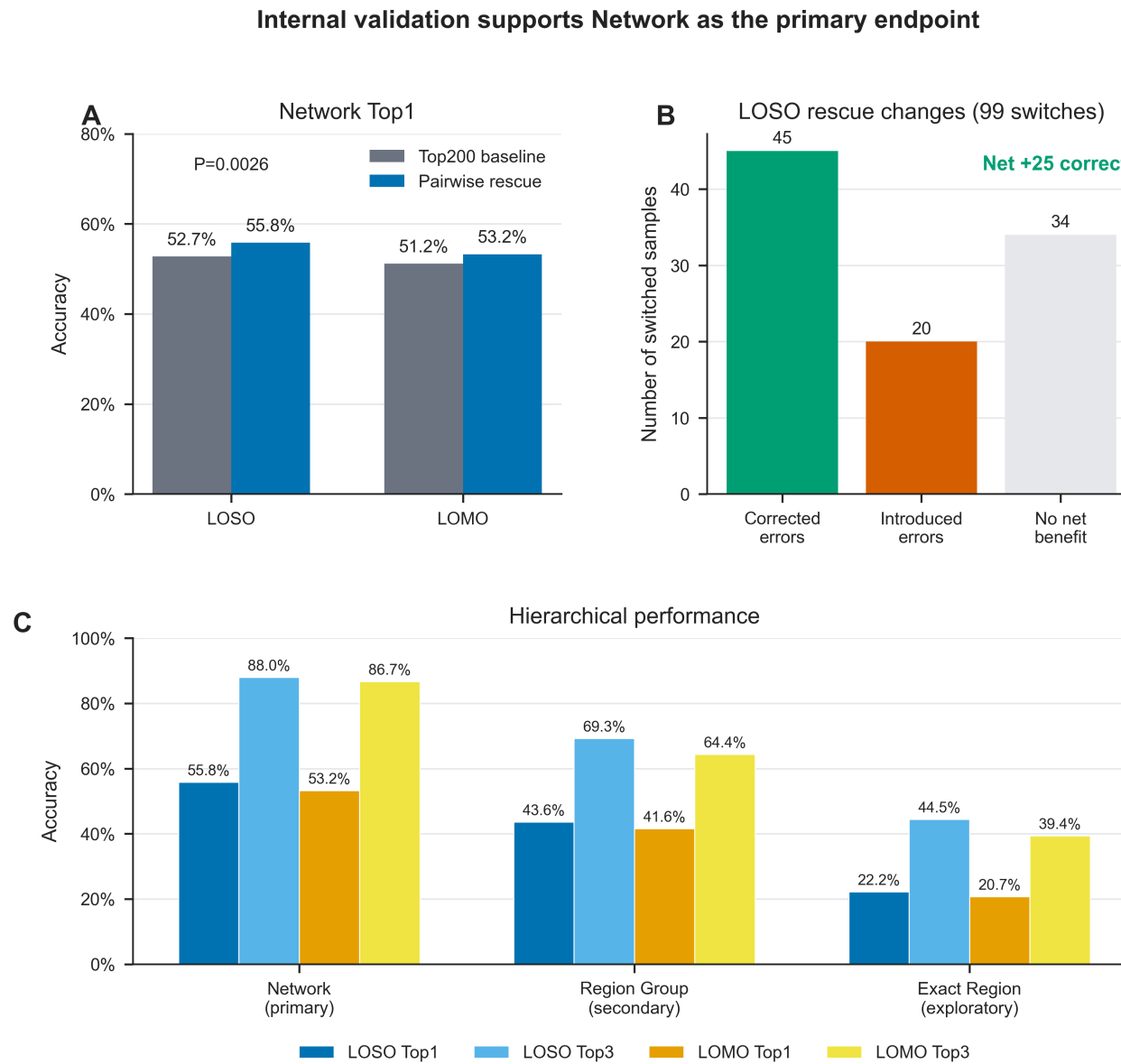
Bioinformatics Application Note补充图表

cfRNA-BrainTrace Supplementary Figures and Tables

包含Figure S1-S5及Table S1-S5

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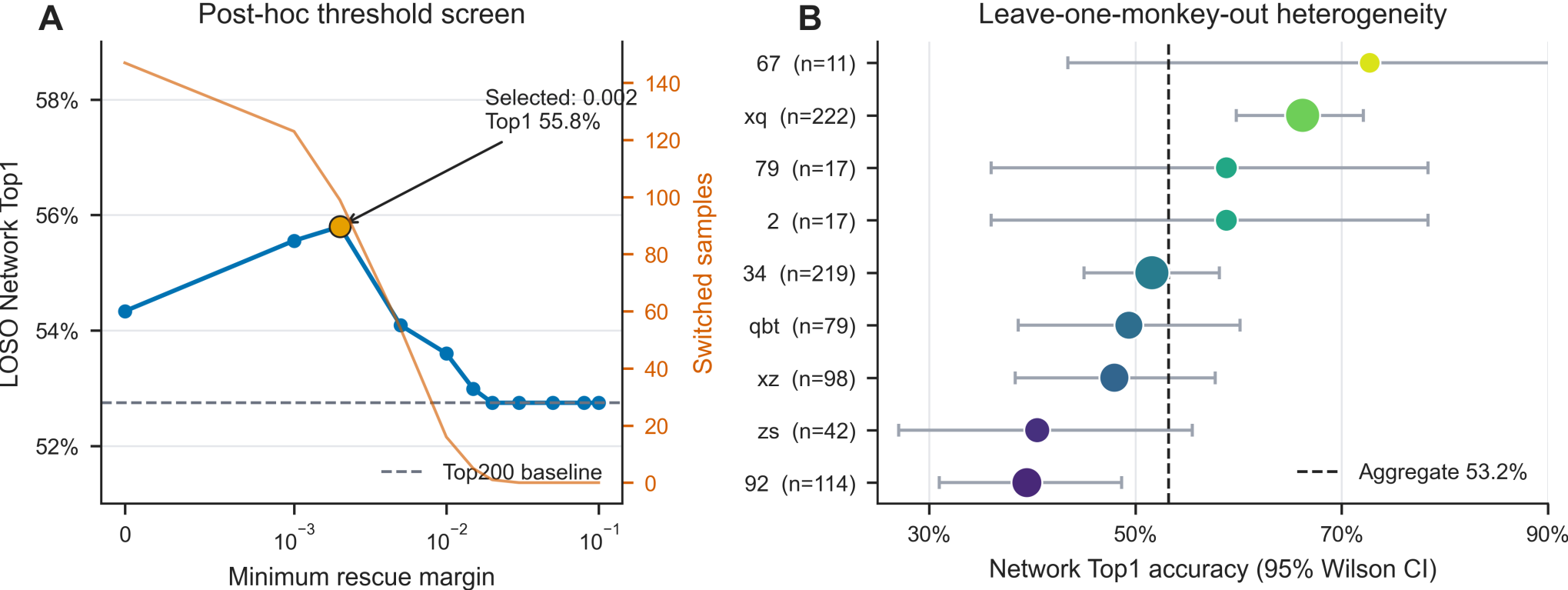
Figure S1. Internal validation across anatomical resolutions



LOSO and leave-one-monkey-out performance at Network, Region Group and Exact Region levels.

Figure S2. Pairwise-rescue threshold and donor-isolated validation

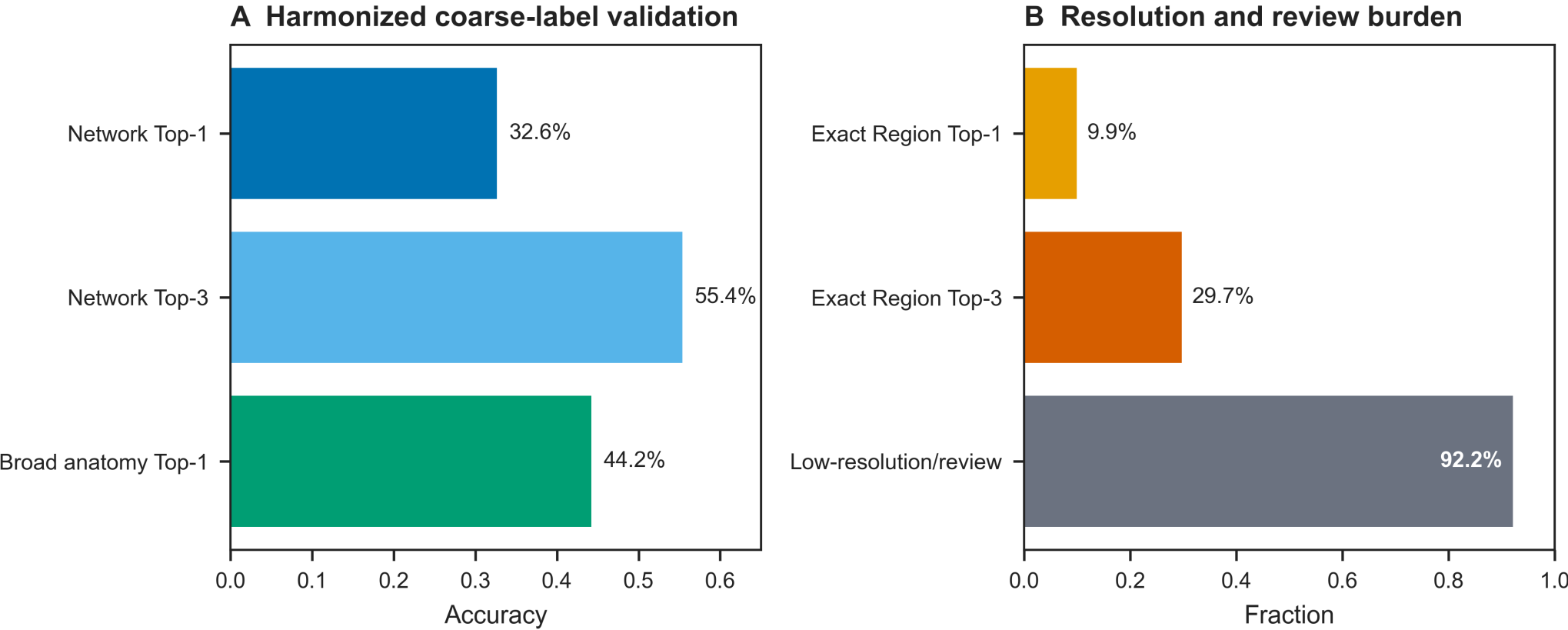
Threshold selection and cross-individual robustness



Retrospective margin-threshold audit and leave-one-monkey-out generalization.

Figure S3. AHBA cross-species validation

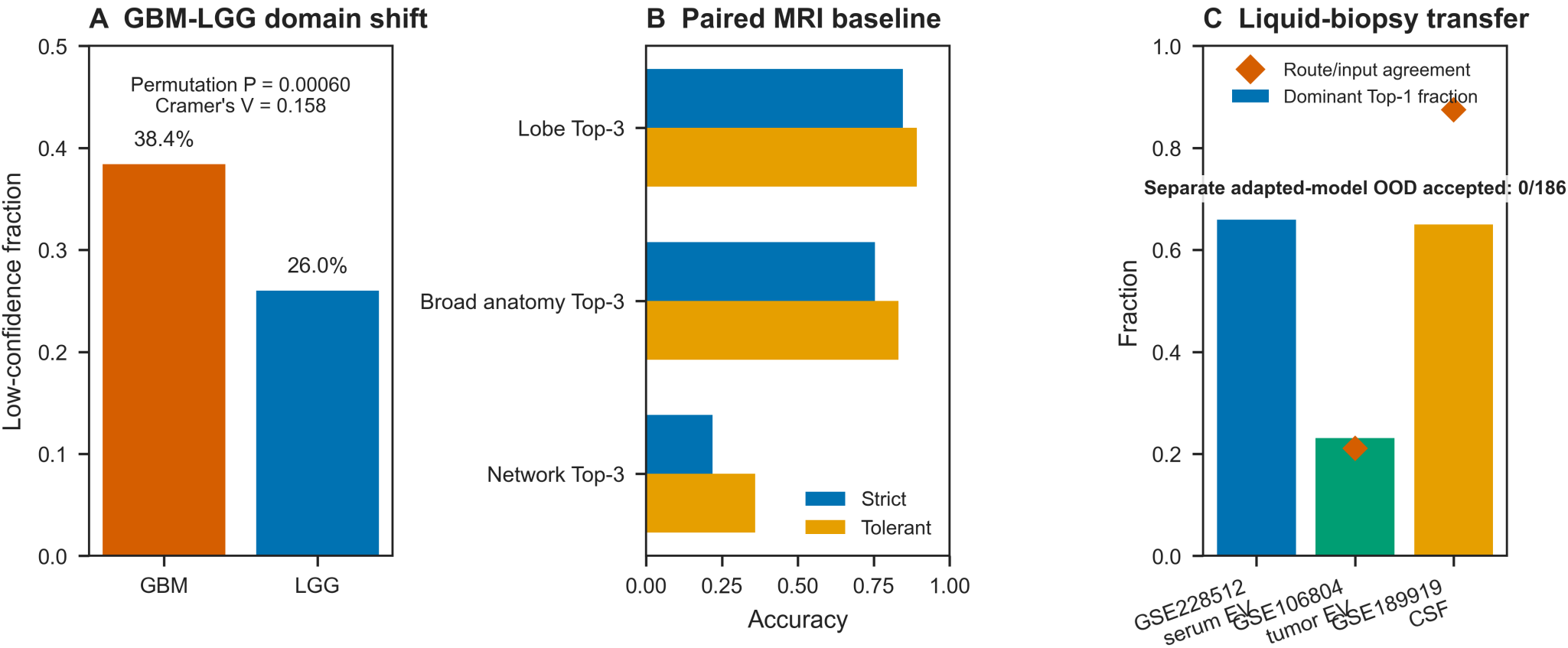
AHBA cross-species validation



Harmonized coarse-label performance and the limitations of exact-region transfer.

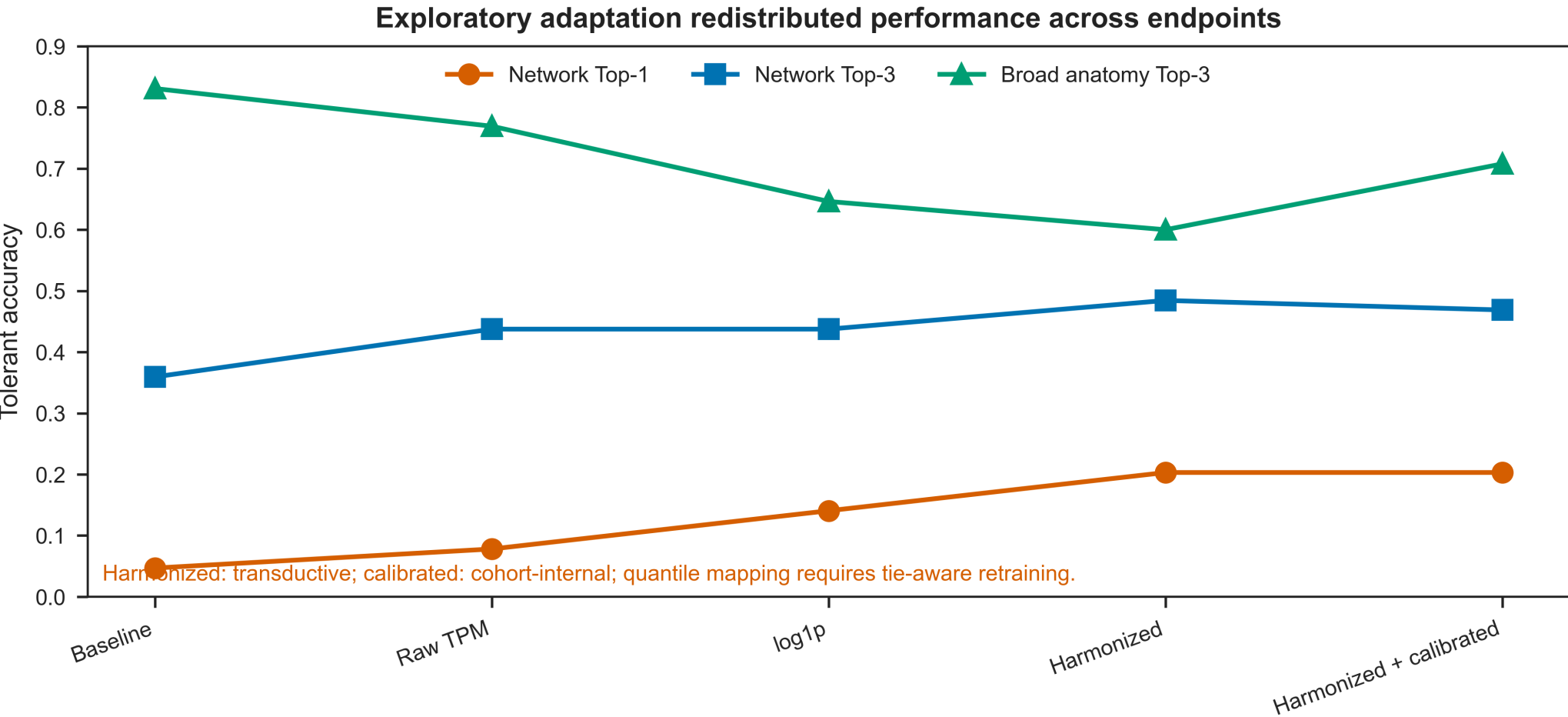
Figure S4. Glioma and liquid-biopsy transfer diagnostics

Glioma validation and liquid-biopsy transfer limits



Paired glioma transcriptome-MRI results and liquid-biopsy transfer limits. Biofluid cohorts lacked imaging truth.

Figure S5. Exploratory adaptation sensitivity



Endpoint-dependent adaptation results. Adapted routes are supplementary and are not part of the locked workflow.

Table S1. Datasets and analysis roles

Dataset	Material	Evaluable size	Label space	Role
Bo2023 macaque atlas	Macaque brain tissue RNA-seq	819 samples; 9 monkeys	110 regions; 10 Networks	Model development; LOSO and donor-isolated validation
AHBA	Normal human brain RNA-seq	242 total; 233 supported	Harmonized Network/broad anatomy; 91 exact-mapped	Cross-species external validation
TCGA GBM/LGG	Human glioma bulk RNA-seq	800 patients	GBM and LGG projects	Unpaired disease-domain shift analysis
TCGA-LGG + BraTS	Paired tumor RNA-seq and MRI	65 patients; Network n=64	Direct tumor-mask overlap with TZO116 labels	Paired anatomical validation
Ivy GAP	Glioblastoma microanatomic RNA-seq	122 samples	Five tumor structures	Exploratory disease-domain characterization
GSE228512	Serum extracellular-vesicle RNA	85 GBM; 31 healthy	No patient-level anatomical truth	Liquid-biopsy transfer stress test
GSE106804	Tumor-associated EV capture RNA	13 GBM; 6 healthy	No patient-level anatomical truth	Enrichment and transfer stress test
GSE189919	Cerebrospinal-fluid RNA	40 medulloblastoma; 11 normal	No patient-level anatomical truth	Technical validation and algorithm audit

Table S2. Internal validation metrics

Validation	Endpoint	Analysis role	Top-1 hits/n	Top-1 accuracy (95% CI)	Top-3 hits/n	Top-3 accuracy (95% CI)
LOSO	Network	Primary	457/819	55.8% (52.4-59.2)	721/819	88.0% (85.6-90.1)
LOSO	Region Group	Secondary	355/814	43.6% (40.2-47.0)	564/814	69.3% (66.0-72.4)
LOSO	Exact Region	Exploratory	181/814	22.2% (19.5-25.2)	362/814	44.5% (41.1-47.9)
LOMO	Network	Primary	436/819	53.2% (49.8-56.6)	710/819	86.7% (84.2-88.8)
LOMO	Region Group	Secondary	338/812	41.6% (38.3-45.0)	523/812	64.4% (61.1-67.6)
LOMO	Exact Region	Exploratory	168/812	20.7% (18.0-23.6)	320/812	39.4% (36.1-42.8)

Table S3. External validation metrics

Cohort	Metric	Hits/n	Accuracy (95% CI)	Interpretation
AHBA	Network Top-1	76/233	32.6% (26.9-38.9)	Latest 2026-06-13 summary; primary endpoint
AHBA	Network Top-3	129/233	55.4% (48.9-61.6)	Harmonized cross-species labels
AHBA	Broad anatomy Top-1	103/233	44.2% (38.0-50.6)	Supportive coarse-anatomy endpoint
AHBA	Exact Region Top-1	9/91	9.9% (5.3-17.7)	Exploratory exact-mapped subset
AHBA	Exact Region Top-3	27/91	29.7% (21.3-39.7)	Exploratory exact-mapped subset
TCGA-LGG/BraTS	Lobe Top-3 strict	55/65	84.6% (73.9-91.4)	Supportive paired validation
TCGA-LGG/BraTS	Lobe Top-3 tolerant	58/65	89.2% (79.4-94.7)	Supportive paired validation
TCGA-LGG/BraTS	Broad anatomy Top-3 strict	49/65	75.4% (63.7-84.2)	Principal disease-scenario readout
TCGA-LGG/BraTS	Broad anatomy Top-3 tolerant	54/65	83.1% (72.2-90.3)	Principal disease-scenario readout
TCGA-LGG/BraTS	Network Top-3 strict	14/64	21.9% (13.5-33.4)	Secondary approximation
TCGA-LGG/BraTS	Network Top-3 tolerant	23/64	35.9% (25.3-48.2)	Secondary approximation

Table S4. Domain-adaptation sensitivity

Route	Analysis role	Uses target unlabeled distribution	Uses paired labels internally	Network Top-1 tolerant	Network Top-3 tolerant	Broad Top-3 tolerant
Baseline	Locked	No	No	4.69%	35.94%	83.08%
Adapted raw TPM	Exploratory sensitivity	No	No	7.81%	43.75%	76.92%
Adapted log1p	Exploratory sensitivity	No	No	14.06%	43.75%	64.62%
Adapted harmonized	Transductive sensitivity; tie-aware retraining required	Yes	No	20.31%	48.44%	60.00%
Harmonized + calibrated	Cohort-internal experiment; tie-aware retraining required	Yes	Yes	20.31%	46.88%	70.77%

Table S5. Liquid-biopsy transfer stress tests

Cohort	Material	Groups	Baseline dominant Top-1 fraction	Separate adapted-model OOD accepted	Interpretation
GSE228512	Serum EV RNA	85 GBM; 31 healthy	65.9% GBM; 71.0% healthy	0/116	High entropy and near-uniform Top-1 probability; no localization accuracy
GSE106804	Tumor-associated EV RNA	13 GBM; 6 healthy	23.1% in GBM	0/19	Lower class dominance but lower margin and 21.1% adapted-route agreement
GSE189919	CSF RNA	40 medulloblastoma; 11 normal	65.0% MB; 45.5% normal	0/51	178/200 markers; 67/67 audit checks; Top1 distribution $P=0.178$; baseline TPM-CPM agreement 87.5%/90.9%